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Towards knowledge graph reasoning for supply chain risk management using graph neural networks

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ABSTRACT

Modern supply chains are complex, interconnected systems that contain emergent, invisible dependencies. Lack of visibility often hinders effective risk planning and results in delayed discovery of supply chain problems, with examples ranging from product contamination, unsustainable production practices, or exposure to suppliers clustered in geographical areas prone to natural or man-made disasters. Initiatives that rely on manual collection of data often fail due to supply chain complexity and unwillingness of suppliers to share data. In this paper, we propose a neurosymbolic machine learning technique to proactively uncover hidden risks in supply chains and discover new information. Our method uses a combination of graph neural networks and knowledge graph reasoning. Unlike existing research our model is able to infer multiple types of hidden relationship risks, presenting a step change in automated supply chain surveillance. The approach has been tested on two empirical datasets from the automotive and energy industries, illustrating that it can provide inference in multiple types of links such as companies, products, production capabilities, certifications; thereby facilitating complex queries that go beyond who-supplies-whom. As such, additional risk insights can emerge from graph structure, providing practitioners with new knowledge.

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1. Introduction

With globalisation and outsourcing trends, modern supply chains have become interconnected and complex (Christopher and Holweg 2011a). Companies typically are aware of actors in their first two tiers, but gradually lose visibility further downstream (Caridi et al. 2010). Managing risk that arises from the interconnections of actors has been challenging due to the lack of incentives for companies to share information on whom do they buy from and supply to, what ingredients are used in their products, and information relating to product quality, ethics and sustainability. This lack of visibility has rendered risk managers with a limited view of the supply chain risk that their companies are exposed to (Caridi et al. 2014). Unknown risk exposure may be damaging to a company's reputation (for example, through the discovery of unsustainable or unethical downstream production practices), as well as operational performance (for example, through disruptions that cascade through the chain of dependencies Dolgui, Ivanov, and Sokolov 2018).

To address these challenges, the last two decades have seen the development of a large body of literature on Supply Chain Risk Management (SCRM) (Ho et al. 2015). Our contribution within this body of literature centres around improving supply chain visibility. Several techniques have been proposed to improve the visibility of hidden relationships in supply chain (Wichmann et al. 2020) such as sending questionnaires to firms, using intermediaries who perform supply chain surveillance, deploying tracking technology including RFID.

A persistent challenge within these approaches is that they still require companies to share their information, of which accuracy and completeness cannot be guaranteed. It is entirely possible that suppliers do not disclose strategically important relations, operational and environmental practices due to a lack of trust (Collier and Sarkis 2021). Hence the information one can obtain using the above techniques, whilst useful, is not complete. There is a need for statistical techniques that can gather and further infer unseen information by leveraging existing knowledge. To this end, the creation and use of

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automated, Artificial Intelligence (AI) based approaches has been recently proposed. This is termed as 'Digital Supply Chain Risk Surveillance (DSCRS)', more specifically defined as the proactive monitoring of digital data that allows firms to track, manage and analyse information related to a supply chain network for the management of risk (Brintrup et al. 2022). DSCRS examples include the use of natural language processing in extracting supply chain maps from publicly available data on the web using Natural Language Processing (Wichmann et al. 2020), use of Twitter to monitor supply chain disruptions (O'Leary 2015), predicting possible delays and hidden links (Brintrup et al. 2018; Kosasih and Brintrup 2021). Much of this literature has emerged after the wider spread use of AI in supply chain enabled existing data from distributed sources to be analysed more effectively (Baryannis et al. 2019; Brintrup 2020; Pournader et al. 2021; Rodríguez-Espíndola et al. 2020).

In this paper, we contribute to this emerging literature by developing and empirically testing a novel method that advances the state of the art in the use of DSCRS. In the existing literature, AI-based link prediction methods have been used to infer hidden supplier–buyer relationships (Kosasih and Brintrup 2021; Brintrup et al. 2018). However, hidden risks can extend to other types of information. For example, a buyer may be interested in whether a product contains certain allergens or controversial materials (such as poor supplier quality in agri-food supply chain (Zhao et al. 2020)), whether a company has a production plant that is located in climate change or disaster prone area (Er Kara, Ghadge, and Bititci 2021), whether a certain upstream supplier has Environmental and Social Governance (ESG) certification, or whether a supplier is owned by unlawful entities and many others (Ho et al. 2015). In this paper, we demonstrate that such hidden, interlinked information can be represented and inferred by constructing a 'Supply Chain Knowledge Graph (SC-KG)'. A SC-KG illustrates the relationships between different entities as a supply chain network and allows additional, contextual knowledge to arise through relationship structures.

Constructing a KG is especially useful for supply chains because it makes multi-type link prediction possible, by discovering hidden information via complex queries that extend beyond who-supplies-whom. Data that are represented via graphs give us additional, structural features to train Machine Learning algorithms, which have been particularly successful in cases where frequentist algorithms such as Neural Networks underperform (Zhou et al. 2019).

Our proposed method is a based on a neurosymbolic framework that combines graph neural network (GNN) and knowledge graph (KG). KG can bring data from multiple sources together and represent them as a graph structure. It also enables complex multi-hop queries to be made, such as 'which gear manufacturer supplies to one of my suppliers?'. Combining a KG with a GNN allows the prediction of multiple types of hidden links in supply chains, making the information more complete. For example, the graph neighbourhood structure of a gear producer would inform the prediction of whether a link to a said manufacturer may exist.

We have tested SC-KG on datasets from two sectors: automotive and energy, which collectively include over 40,000 UK and internationally based companies. Our results demonstrate that we could infer hidden links across different types of relationships, facilitating complex risk queries for practitioners.

This paper is organised as follows. Section 2 discusses existing literature on supply chain risk management (SCRM) and reasoning with AI. Section 3 discusses how typical queries in SCRM could be represented as a reasoning problem and outlines the SC-KG mathematical framework. Section 4 details the methodology, and Section 5 explores model performance with two supply chain datasets. Section 6 provides discusses results and Section 7 provides conclusions and managerial insights.

2. Literature review

2.1. Supply chain risk management

Modern supply chains are interconnected and complex, resulting in a multitude of risks, the exposure to which might be unknown or cannot be estimated (Christopher and Holweg 2011a).

Supply Chain Risk Management (SCRM) has become an important field that studies methods to manage such risks better. Through the use of empirical and model-based studies, SCRM literature offers practitioners guidelines for implementing reactive and proactive risk management strategies (Tomlin 2006). While proactive strategies include safety stock, multi-sourcing, information sharing, supplier collaboration, and contingent rerouting, reactive strategies may include business continuity planning, and expediting late orders (Ho et al. 2015).

Contemporary research in SCRM has highlighted the risk that arises from emergent, invisible parts of the supply chain motivated by natural disasters such as the 2011 Tohuku earthquake and Thailand Floods, Covid-19, and man-made disasters such as the 2018 Michigan fire, 2021 Suez Canal blockage (see Jüttner and Maklan 2011; Ho et al. 2015; Ivanov et al. 2019). Reputational risk cases where suppliers are involved in deforestation, conflict minerals and supply chain slavery, also serve among prime motivators for companies' willingness to discover and manage hidden risks (Busse et al. 2017).

Before any proactive risk management strategy can be modelled and implemented, potential risks need to be mapped. For any risks to be mapped, information on the network needs to be more complete. However, risk monitoring in supply chains remains an understudied area (Ho et al. 2015). In this work, we focus on the improvement of supply chain visibility, so that proactive SCRM can be practiced on emergent parts of the supply chain that lay beyond the control of the buying firm. While a multitude of approaches has been proposed to improve SC visibility, our focus is specifically on AI-driven methods that do not rely on data sharing.

Recently, researchers have highlighted the promise of Artificial Intelligence in SCRM (Baryannis et al. 2019; Brintrup 2020) and more specifically in obtaining more information on supply chains. Wichmann et al. (2020) proposed natural language processing (NLP) approaches for automated extraction of supply chain maps from World Wide Web. Here, after entity recognition, NLP classifiers are trained recognising supplier–buyer relationships. Brintrup et al. (2018) suggested a link prediction framework to infer supplier–buyer relationships using a complex network based approach, where possible supplier competition, and product dependencies are identified through graph mining, which then formed a feature set to train classifiers. Kosasih and Brintrup (2021) trialled a graph neural network approach to train classifiers using only topological data, which significantly improved upon previous approaches in terms of accuracy and completeness. However, all approaches proposed focused on a single relationship type, that is between buyer-suppliers. Although these works advance the state of the art by providing a tool to detect potential hidden relations in a buyer's supply chain, more complete information could yield more effective SCRM practice. To illustrate this point, a number of exemplary practitioner queries that have been raised in literature are shown in Table 1.

The first query, would require not only that a given supplier–buyer relationship exists, but also the product that is being procured, which in this case is a food product consisting of many ingredients. To address such a query the existence of multiple relationships, in the form of 'supplier–buyer', 'supplier-product', and 'product-ingredient', need to be predicted. Similarly, the query posed by Suraraksa and Shin (2019) points to the need for ' supplier–buyer' and 'supplier-certification' relationships. Hence, while ' supplier–buyer' relationship detection is a first step to detecting risk, additional information may help answer more complex risk queries, making the analysis finer-grained. The manual obtainment of such data would be a tremendous task in complex, large-scale supply networks. Further, relationships might not be certain and may need to be inferred from data. For instance, we might know certain ingredients tend to be present in certain products types (e.g. 'a cake may have eggs'), but this may not always be guaranteed (e.g. 'vegan cake'), or certain ingredients may be 'hidden' (e.g. the use of the term 'vegetable oil', instead of 'palm oil'). In the next section, we review how the field of AI, specifically, *Reasoning*, may help address these challenges.

Table 1. Risk factors.

Risk query	Reference
Does my supplier's product contain hidden allergens?	Zhou et al. (2019), Ma et al. (2021)
Is my supplier connected to this company?	Brintrup et al. (2018), Kosasih and Brintrup (2021)
Is my supplier's raw material producer company compliant with international quality/sustainability/governance standards?	Suraraksa and Shin (2019), Giannakis et al. (2020)
Is my supplier owned by a hidden offshore tax-sheltered firm?	Zhuhadar and Ciampa (2021), Srivastava and Singh (2018)
Is my supplier's contract manufacturer located in a certain risky geography?	Ledwoch et al. (2018), Sodhi (2014)

To summarise, we have identified two research gaps in the literature for improving supply chain risk visibility. First, existing techniques such as RFID, NLP and questionnaires rely on companies' willingness to share their information. However, the completeness of information that one can obtain from these approaches cannot be guaranteed. Hence, an automated method that can infer hidden information from these data would be useful to close gaps in knowledge. Second, previously proposed AI-based methods have focused only on predicting hidden relationships between companies, which is a single type of risk, despite the existence of multiple types of risk in supply chains. Therefore, there is a gap in techniques that could identify multiple type of risks simultaneously.

2.2. Reasoning with AI

Inferring new knowledge such as hidden relationships from existing information has also been studied in the sub-field of Artificial Intelligence called 'Reasoning'. While reasoning has been studied in other domains, recent surveys found that its use in supply chain management remains scarce (Baryannis et al. 2019; Brintrup 2020).

Two approaches to perform reasoning in AI exist: symbolic and connectionist (Russell et al. 2010). Symbolic models are built on first-order logic to derive new information from the existing set of facts stored in a knowledge base. This class of algorithms is inherently interpretable as one can trace a finding back to the

original set of knowledge that was used by the model to derive a conclusion. For instance, consider the logic statement 'UPS uses GE Aviation's product'. To consider whether the statement is true, a symbolic model would be needed to trace the existence of paths in a set of facts in a knowledge base. If one knows that 'GE Aviation supplies to Boeing' (*supplies_to (GE Aviation, Boeing)*) and 'Boeing supplies to UPS' (*supplies_to (Boeing, UPS)*), then with certain assumptions one can reason that 'UPS is a customer of GE Aviation' (*is_customer (UPS, GE Aviation)*). One can thus interpret how this decision is made by tracing back to the original two facts that were used in the reasoning process. However, while symbolic approaches have the advantage of interpretability, constructing facts in knowledge base is quite laborious. Such logic reasoning approach also requires a knowledge base to be comprehensive, since if any single fact is missing (i.e. *supplies_to (Boeing, UPS)*), it can not derive conclusions (i.e. *is_customer (UPS, GE Aviation)*).

On the other hand, connectionist models are good at extracting information from data automatically and representing them in data structures that can be manipulated by arithmetics. The connectionist paradigm, often referred to as neural networks or deep learning, helps derive functions that can learn to map objects (texts, images, graphs, manifolds and other groups) to vectors in Euclidean space (embeddings). The parameters of the function are fitted to training data such that the embedding space captures some latent information of the input objects. For instance, Transformer Network (which can be generalised as a Graph Neural Network) can capture complex relationships of words in natural language through attention weighting and recurrent layers. To illustrate how a connectionist model can be used for reasoning, consider feeding a news about UPS ordering nineteen Boeing 767 Freighters as their e-commerce business expands (Reuters 2021), to a Transformer Network. After being trained on many text to extract supplier–buyer relationship (such as Wichmann et al. 2020), the model might identify that there are Boeing, UPS and a transaction happening between them in the text. Based on training data, it might derive that Boeing must be supplying to UPS from the article (*supplies_to (Boeing, UPS)*). It might, however, not be clear why the model could understand that the word *order* means purchase and not giving command, although this would be obvious to a human. While explainability has been explored, deducing the correctness of the explanation is still a challenge (Das and Rad 2020). In fact, this is why connectionist models often remain as a black box, as it is difficult to trace through the fitted function's weights of what it has learned.

Researchers have been trying to combine both symbolic and connectionist models to leverage on each others' strengths (Russell et al. 2010). The objective is to create a model that can automatically extract information automatically from data and to reason over them so as to derive new knowledge. Continuing our example, connectionist models can be used to extract information from data (i.e. *supplies_to (Boeing, UPS)* from a news text in the internet), while symbolic models can be used to reason over the set of extracted information (i.e. *supplies_to (GE Aviation, Boeing)* and *supplies_to (Boeing, UPS)*) to derive knowledge (*is_customer (UPS, GE Aviation)*). In AI, this framework is called a 'neurosymbolic model'. A recent survey (Garcez and Lamb 2020) has identified six type of neurosymbolic models. Type 1 is standard deep learning, where inputs and outputs are symbols but the operations are carried out with a neural network. Type 2 utilises a neural network as the core system, but it is loosely coupled with a symbolic solver such as Monte Carlo Tree Search (AlphaGo). Type 3 uses a neural network for a task (such as objection detection) and then interacts with a pure symbolic system for another complementary task (such as query answering). Examples are Concept Learner and deepProbLog (Manhaeve et al. 2018). Type 4 compiles symbolic knowledge into a neural network. Some examples include Logical Neural Networks, ANFIS. Type 5 maps symbolic logic rule as embeddings and that acts as a regularizer as the neural network's loss function (also called as tensorization-based models). Examples include Logic Tensor Networks and Tensor Product Representations (Badreddine et al. 2022). Recent works in this space include Graph Neural Networks-based embeddings (Hamilton 2020). Lastly, Type 6 is the true symbolic reasoning inside neural engine. A full-fledged Type 6 system does not exist yet (Garcez and Lamb 2020).

In this paper, we develop a Type 5 neurosymbolic model, where we use GNN to extract information from data as embeddings and use symbolic logic to infer the existence of new knowledge in Supply Chains. We choose Type 5 as this is the state of the art neurosymbolic model in the current literature.

3. Knowledge graph reasoning

In this section, we demonstrate how Knowledge Graph Reasoning (KGR) can be applied to SCRM. There are four steps to perform KGR: (1) Data Collection, (2) Ontology Engineering, (3) Knowledge Graph Creation and (4) Knowledge Graph Reasoning.

The first step involves the collection of data pertaining to the supply chain risk management task. Practitioners

need to identify which data types and sources that will be most useful for SCRM tasks. These are typically traditionally stored in multiple relational databases in silo across different departments and may need to be combined with externally accessed datasets (such as financial information, industrial databases). Therefore, answering queries such as those from Table 1 requires practitioners to query different databases such as supplier management, bill of materials, certification compliance, etc. In order to break the silo, a particular representation called Knowledge Graph (KG) has been proposed (Aasman 2017). KG provides a unified view of the databases as each previously siloed information is collected as a connected network of data.

Before a KG can be built an *ontology* must be designed. The ontology defines the different data types and attributes, and enlists a taxonomy of items that are present in the data. Once the ontology is designed, we proceed to creating the KG out of the data entries, based on the ontology. Each item in the data is assigned to entities (also called nodes in a graph), and the relationships between data are represented as edges in the graph, based on the ontology definition.

Once a KG is built, we can perform reasoning over it. Here, we use a neural network-based model to derive hidden facts by predicting missing links and add them in the graph so knowledge gaps are closed. Then, we convert risk queries into predicate logic statements and check if the answer to such question can be found in the enriched KG.

We describe the construction and reasoning of KG in more detail below.

3.1. Knowledge graph

A KG is represented by a heterogeneous graph $\mathcal{G}$ and its ontology $\mathcal{O}$, the former being the actual data and the latter being its schema. KG can also be seen as a collection of facts represented by predicate logic statements. These components are defined as follows.

Heterogeneous graph: $G = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ is the set of entities/nodes (both used interchangeably), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of relations. We can also refer to each pair of entity $v_1, v_2 \in \mathcal{V}$ and the relation between them $\epsilon \in \mathcal{E}$ as a triplet (v_1, ϵ, v_2). Triplets are used in Predicate Logic.

Ontology: $\mathcal{O} = \{u \in \mathcal{V} \mapsto \Lambda, (\epsilon_1, \epsilon_2) \in \Lambda \times \Lambda \mapsto \mathcal{R}_\epsilon \in \mathcal{R}\}$ where Λ is the set of entity types and $\mathcal{R}$ is the set of relation types. Ontology is a function that maps each entity in heterogeneous graph $u \in \mathcal{V}$ to a particular entity type Λ, and pair of entity types $(\epsilon_1, \epsilon_2) \in \Lambda \times \Lambda$ to possible relation types $\mathcal{R}_\epsilon$. Ontology is analogous to a schema of the database. In a supply chain, entity types may represent objects such as companies, products, certifications, countries, and other discrete entities that exist in the network. Meanwhile, relation types represent connections between these discrete entities.

Table 2. Risk factors as logic statements.

Risk query	Logic statement
Does my supplier's product contain hidden allergens?	Has_Ingredient (product, nuts)
Is my supplier connected to another company?	Supplies_to (company X, company Y)
Is my supplier's raw material producer company compliant with international quality/sustainability/governance standards?	Has_Certificate (company X, ISO9000)
Is my supplier owned by a hidden offshore tax-sheltered firm?	Is_Owner(firm X, company Y)
Is my supplier's contract manufacturer located in a certain risky geography?	Is_Located (company X, region Y)

Predicate logic: The set of all possible pairs of nodes and the relations that connect them are called triplets ((v_1, ϵ, v_2) for all $v_1, v_2 \in \mathcal{V}$ and $\epsilon \in \mathcal{E}$). We could also view triplet (v_1, ϵ, v_2) as a predicate logic statement $r_\epsilon(v_1, v_2)$ with a satisfiability value of True or False. For instance, a product type 'cake' containing the ingredient 'nuts' can be represented as a triplet containing two nodes (cake and nuts), and one relation type (has_ingredient). This predicate logic statement will be *has_ ingredient* (cake, nuts). If the satisfiability of this statement is True, that means a given cake instance really contains nuts, while False means nuts do not exist in the given cake. In a theoretical fully connected graph, all triplets will have satisfiability of True, meaning that the statement is true since all triplets exist in the KG. However, in sparse graphs, some triplets can have satisfiability of False, meaning that such a relationship among that pair of nodes does not exist. Some triplets are False because they really do not exist in the real world, while some are because of the missing information in the input data due to lack of visibility problem.

The task of KG reasoning is to infer which of the False triplets are due to missing information in the input data, and they are supposed to be True in reality. Therefore, users can use KG reasoning capability to answer their questions. For instance, using the same example as before, if we would like to know whether a certain product cake contains the ingredient nuts, we can pose the query as a predicate logic statement has_ingredient (cake, nuts). We then perform KG reasoning to find out if it is True or False. Similarly, we could represent all risk queries in Table 1 as their respective predicate logic statements shown in Table 2.

Determining satisfiability of a given triplet is closely related to another machine learning task called *link prediction*. We shall review the concepts on link prediction in the next section, leading to the concept of KG reasoning.

3.2. Link prediction in homogeneous graph

As mentioned before, determining the satisfiability of predicate logic statements derived from KG triplets is closely related to another machine learning task called link prediction. The term 'Link prediction' is mainly used to refer to predictive tasks in homogeneous graphs, where there is only single entity type and relation type, hence all nodes and edges are of the same types. Link Prediction is used in many domains such as social network analysis, recommendation system or e-commerce. Main types of link prediction algorithms are discussed below.

Similarity-based techniques measure similarity between two nodes to predict if there are links between them. The calculation of similarity is context dependent and may include counting the number of common neighbours between nodes, node degree and other formulae. Examples of similarity-based algorithms include Jaccard Coefficient (Liben-Nowell and Kleinberg 2007), Katz (Katz 1953), LHN Index (Leicht, Holme, and Newman 2006), Preferential Attachment (Barabási and Albert 1999), Adamic-Adar (Adamic and Adar 2003), Resource Allocation (Zhou, Lü, and Zhang 2009), path-based similarity (Lü, Jin, and Zhou 2009) and Bayesian estimation (Liu et al. 2011).

Topology-based techniques have also been developed. These algorithms use probability models that are trained using maximum likelihood estimation to produce a given network topology (using structural properties such as the existence of a hierarchy, communities, large hubs or other motifs). The model could then be used to infer missing links in the structure. An example of such algorithm is hierarchical random graph model proposed by Clauset, Moore, and Newman (2008). However, several previous studies (Hearnshaw and Wilson 2013; Thadakamaila et al. 2004; Brintrup, Ledwoch, and Barros 2016; Brintrup, Wang, and Tiwari 2017) have shown that characterising network topology with manually crafted motifs does not result in reliable statistics of the overall network topology, hence rendering this approach less accurate for supply chain when used as standalone model.

Node feature extraction models use topology-derived attributes from the nodes to predict hidden links. Examples include matrix factorisation (Koren, Bell, and Volinsky 2009), Stochastic Block Model (Airoldi et al. 2008), DeepWalk (Perozzi, Al-Rfou, and Skiena 2014), LINE (Tang et al. 2015) and node2vec (Grover and Leskovec 2016). Particularly in supply chains, SNLP (Brintrup et al. 2018) is an example of this approach. Four handcrafted attributes were manually developed: number of suppliers, number of common buyers, product outsourcing associations and product portfolio overlaps. These attributes are then fed into a binary classifier to predict link existence.

Graph Neural Networks (GNN) based techniques have been shown to outperform many of the previous models in multiple domain, including airline networks, political blogs, citation networks, power grid, *E. coli* metabolites reactions and router-level internet (Hamilton 2020; Bruna et al. 2014; Duvenaud et al. 2015; Kipf and Welling 2017; Niepert, Ahmed, and Kutzkov 2016; Zhang and Chen 2017, 2018; Zhang et al. 2020; Huang and Zitnik 2021; Teru, Denis, and Hamilton 2020). Particularly in supply chains, Kosasih and Brintrup (2021) have shown that GNN outperforms previous link prediction approaches.

Building on the state of the art GNN-based model from link prediction (Kosasih and Brintrup 2021), we propose to use GNN for our KG reasoning model, particularly for the task of predicting satisfiability of given triplets.

3.3. Reasoning in heterogeneous graph

Our KG reasoning model is based on Type 5 neurosymbolic architecture. We use KG and predicate logic statements to compile symbolic logic rules. We then convert these symbolic statements to embeddings using GNN, as it has been shown in homogeneous graphs that GNN can generate good representations that work well for link prediction. Hence, in order to determine satisfiability of a logic statement $r(x, y)$, we use an encoder function to convert the entities to their continuous-valued embeddings (x to $\mathbf{x}$ and y to $\mathbf{y}$) and predicate to a decoder function f_r that outputs a value between 0 (False) and 1 (True). Therefore, answering $r(x, y)$ is done by calculating the output of function $f_r(\mathbf{x}, \mathbf{y})$. Figure 1 illustrates how our model works, with r being supplies_to and f_r being simple dot product to calculate the distance between two embedding vectors.

3.3.1. Encoder

We propose to use heterogeneous graph neural networks (GNN) as the encoder model to produce embeddings for input entities. GNN is a message-passing model that performs three stages of operations (Hamilton 2020)). First, it initialises the embeddings for each node randomly. Second, it executes convolution in a way that is analogous to a Convolutional Neural Network (CNN). The convolution operation aggregates embeddings from each node's neighbourhood into a neighbourhood embedding feature. The main difference between heterogeneous and homogeneous GNN lies here. In homogeneous GNN, there is only one type of edge, hence aggregation can

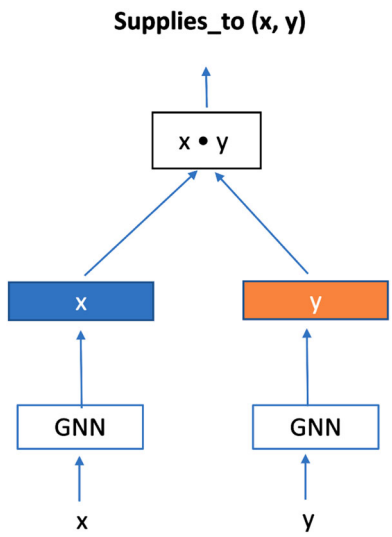


Figure 1. Tensorization-based neurosymbolic reasoning.

be done in one step across all edges. Meanwhile, in heterogeneous GNN, there are multiple types of edges. In practice, we instantiate multiple GNNs, one for each type of edge. Aggregation of information is done with in two stages: aggregate across neighbourhood for each edge type, then sum the aggregated info across node embeddings obtained for each edge. This step is called AGGREGATION. Lastly, we combine the neighbourhood embedding with each node's previous embedding and create a new vector. We repeat the second and third step k-times to obtain k-layer GNN (Teru and Hamilton 2020). This step is called UPDATE. Equation (1) illustrates this process.

$$h_u^{k+1} = \text{UPDATE}^k\left(h_u^k, \sum_{r\in R} \text{AGGREGATE}_r^k(h_v^k, v \in N_r(u))\right) \tag{1}$$

where h_u^0 is initiated randomly. This equation can be implemented in various ways, an example of which is shown in Figure 1. In this equation, each edge type as defined by the ontology, and has its own GNN. The k_{th} GNN layer is parameterised by activation function and weight matrix (σ_r^k, W_r^k).

$$h_u^{k+1} = h_u^k + \sum_{r\in R} \sigma_r^k\left(\sum_{v\in N_r(u)} W_r^k h_v^k\right) \tag{2}$$

where h_u^k, h_v^k are embeddings for node u and v at the k_{th} GNN layer, R is a set of edges defined in the ontology, $N_r(u)$ is the set of neighbours of u that are connected by edge type r.

3.3.2. Decoder

In order to infer that a pair of nodes is connected by an edge, we can define a decoder for each edge type. The decoder takes a pair of node embeddings at the final K_{th} and the edge type (cite DistMult paper).

$$f(u, r, v) = \sigma((h_u^K)^T R_r h_v^K) \tag{3}$$

where R_r is edge type-specific matrix and σ is sigmoid function. In this paper, we set R_r to be an identity matrix, resulting in $f(u, r, v)$ being a dot product between the pair of node embeddings.

3.3.3. Objective function

The model is trained to optimise the satisfiability of the set of triplet facts in the knowledge graph. We frame this as a binary prediction task, where the model needs to predict if a given triplet is True or False. The positive samples (with satisfiability True) are the existing triplets. Meanwhile, the negative samples (with satisfiability False) are randomly sampled from pair of entities without existing edges between them.

$$\mathcal{L} = \sum_{r\in\mathcal{R}} \sum_{(u,v)\in\mathcal{V}} \sum_{y\in\{0,1\}} y \log f(u, r, v) + (1 - y)\log(1 - f(u, r, v)) \tag{4}$$

where y is the label of edge r connecting pair of nodes (u, v) whose values can be either 1 or 0. 1 refers to the existence of an edge (triplet with satisfiability True) and 0 to the lack of an edge (triplet with satisfiability False).

4. Experimental methodology

This section describes how the proposed KG Reasoning for SCRM has been tested with empirical datasets.

Experiments have been performed using two real supply chain datasets to illustrate our approach. The first dataset consists of automotive supply chain data from Marklines, a global automotive industry information provider. This dataset was used in previous link prediction studies from Brintrup et al. (2018) and Kosasih and Brintrup (2021), which allows us to benchmark performance. The second dataset leverages anonymised supply chain data from the energy sector, provided by Achilles Information Ltd, a supply chain risk and performance management company based in the United Kingdom.

For each dataset, we defined an ontology and constructed the Knowledge Graph using the entities and triplets. The triplets were then split into training (70%), validation (10%) and testing (20%). Negative triplets were sampled in order to train the model so as to differentiate between edges and non-edges. The sampling is done such

that in training dataset, there is an equal ratio of positive and negative triplets, while in testing four different ratios were experimented with to stress test the model.

We compared our proposed model with two state-of-the art models from (Brintrup et al. 2018) (SNLP) and (Kosasih and Brintrup 2021) (GNN-LP), as well as random guess as baseline (random-uniform). Following previous literature, the models were evaluated using three metrics: Area Under Receiver Operating Characteristic Curve (AUC), Average Precision (AP) and f1-score. AUC is a score between 0 and 1, and it measures how well the model performs compared to a random classifier. A score of 1 shows that the model is robust. AP is a score between 0 and 1 and it measures the precision vs recall capability of the data. A score of 1 shows that the model has high precision without making any classification mistakes. Meanwhile, an f1-score also ranges between 0 and 1 and is defined as the harmonic mean of precision and recall of prediction result. The closer to 1 the better the model is. Refer to Pedregosa et al. (2011) for more details. Following GNN literature, each experiment has been repeated 10 times from different random initial conditions, in order to obtain the 90% confidence interval.

The core GNN engine is built using a widely used software package called PyTorch Geometric (Fey and Lenssen 2019). Meanwhile, the supply chain dataset is queried from different databases, containing information about different companies, their products, certifications, etc., and how they are dependent with each other. We also obtained the schema and ontology of the supply chain data from the database provider. All experiments were done on a Dell laptop with Intel i9-9980HK 2.4 GHz CPU (16 GB RAM), and NVIDIA GeForce GTX1650 GPU (4 GB RAM).

5. Results

In this section, we present experimental results using two case examples from the Automotive and Energy sectors. First, datasets pertaining to the two use cases are detailed, followed by individual results and discussion on the overall performance of the developed approach.

5.1. Automotive use case

The Automotive dataset is from Marklines, a supplier information company located in Japan. The dataset contains five types of entities: company, country, certification, product and capability. Table 3 shows the number of unique nodes for each entity. Here, a 'Company' entity depicts the legal name of a given company. 'Country' depicts the geographic location of the company entity. 'Certification' refers to the qualifications of a company in a given field, authorised by a recognised legal entity. These could include ISO certifications pertaining the standards that a company complies with. Relations depicted in Figure 2 are those between companies (i.e. supplier–buyer relationships).

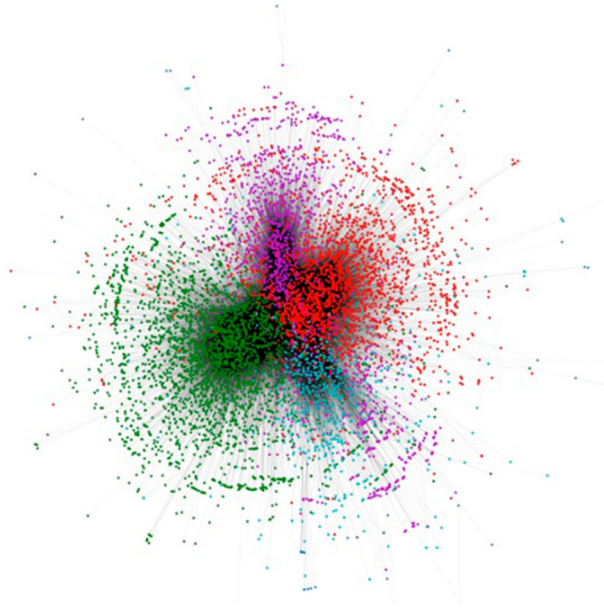


Figure 2. Marklines automotive network visualisation. Only supplier–buyer relations are illustrated. Colors refer to communities in the network, obtained by a greedy community detection algorithm (Clauset, Newman, and Moore 2004).

Table 3. Number of entities.

Entity type	Number of entities
Company	41,986
Country	74
Certification	5
Product	892
Capability	35

Using the aforementioned entity descriptions, we then define the ontology of the automotive dataset as shown in Figure 3. This triplet definition is obtained from the dataset. Using the ontology, a Knowledge Graph can then be populated. Figure 4 shows a snapshot of a subgraph of our supply chain dataset for illustration. As the overall graph is too large to be depicted, only a subset of the triplets is shown.

Following knowledge graph creation, the knowledge graph completion algorithm is then run to find hidden relations on the graph. Properties of the training and testing datasets are shown in Table 4. As it can be seen, the graph has a large number of 'makes product' and 'supplier to' edges, however, 'located in' and 'has certificate' edges are relatively smaller in number. We include both the number of links and negative samples, i.e. 'lack of links'

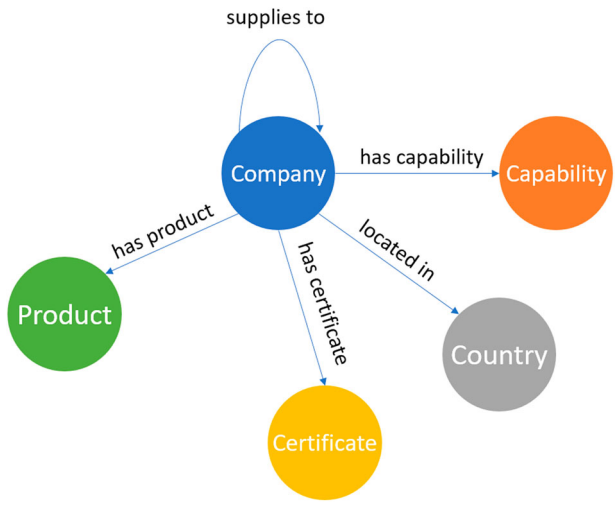


Figure 3. Proposed ontology.

for each edge type on the graph. To test the method's efficacy under imbalanced conditions, the ratio of links to negative samples have been varied in each edge type.

The performance of the model can be seen in Figure 5. The 90% confidence interval is displayed as a shaded area on the figures. We will discuss and compare these later.

5.2. Energy

This section shows how our model performs in the second dataset provided by Achilles Information Ltd, a supply chain risk and performance management company based in the United Kingdom. This is an anonymised supply chain data on energy industry revolving around oil and gas companies in the North Sea, including UK, Norway, Denmark and many others (Figure 6).

The data contain three types of entities: company, country and product as shown in Table 5 and is smaller than the Automotive dataset in both size and edge types.

Similar to the automotive use case, we begin by defining the ontology of this dataset as shown in Figure 7. We can see the connection between company, product and country in the directed graph.

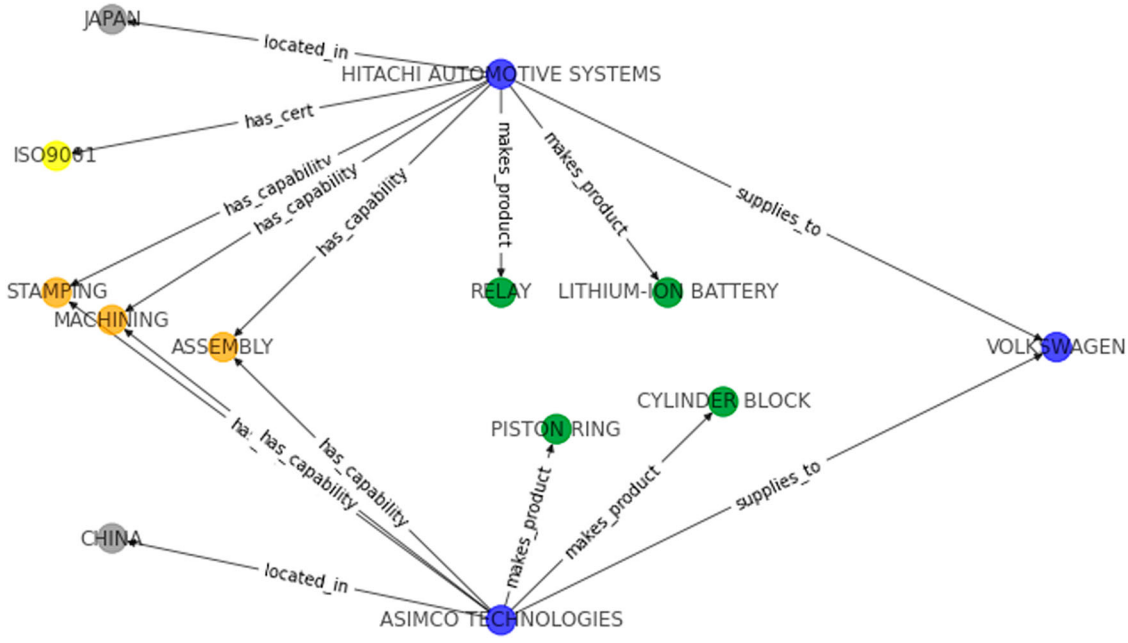


Figure 4. Snapshot of Marklines automotive knowledge graph.

Table 4. Marklines train and test dataset.

Edge type	Links	Negative samples	Ratio
		(a) Marklines Train Dataset	
Supplies to	62,575	62,575	1:1
Located in	28,627	28,627	1:1
Has certificate	22,999	22,999	1:1
Makes product	84,511	84,511	1:1
Has capability	59,291	59,291	1:1
		Marklines Test Dataset	
Supplies to	17,878	17,878/35,756/53,634/71,512	1:1/2/3/4
Located in	8178	8178/16,356/24,534/32,172	1:1/2/3/4
Has certificate	6571	6571/13,142/19,713/26,284	1:1/2/3/4
Makes product	24,145	24,145/48,290/72,435/96,580	1:1/2/3/4
Has capability	16,940	16,940/33,880/50,820/67,760	1:1/2/3/4

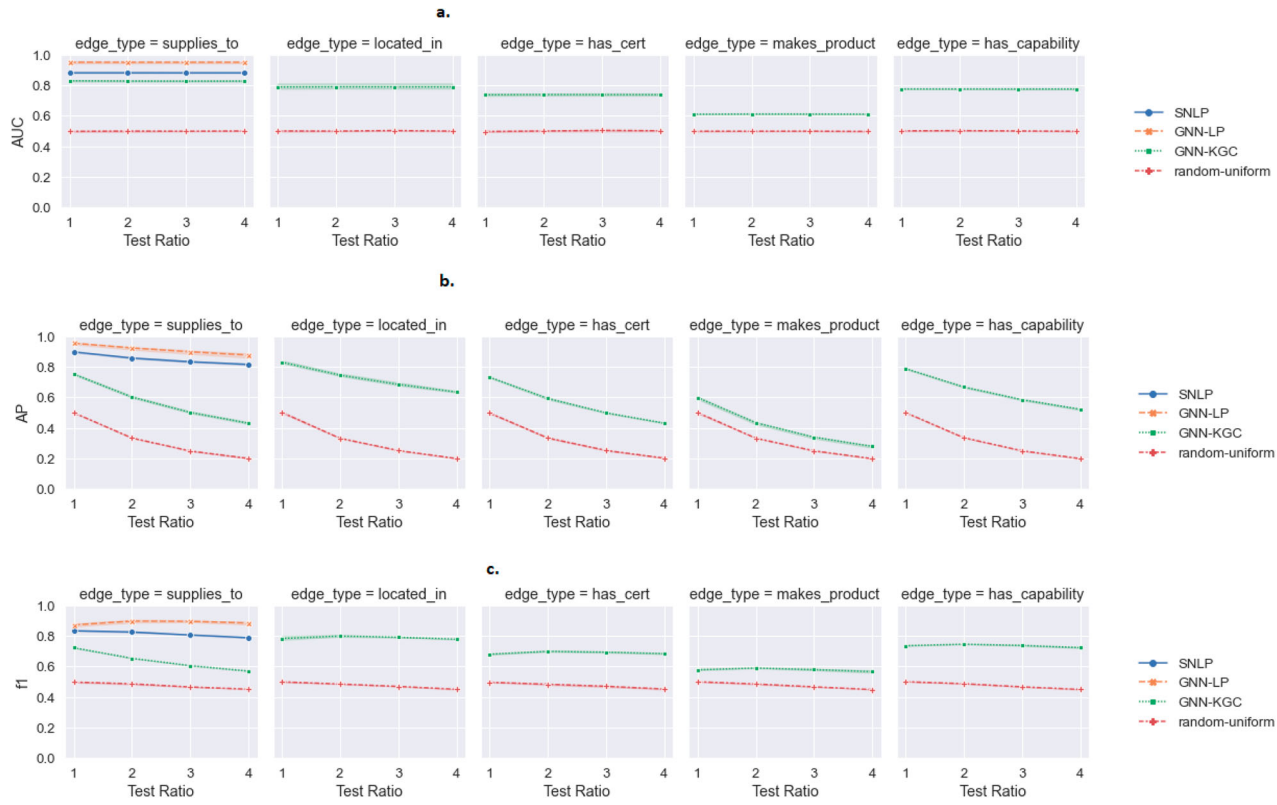


Figure 5. Performance on Marklines dataset. (a) Area under curve across different test ratio, (b) average precision across different test ratio and (c) f1-score across different test ratio.

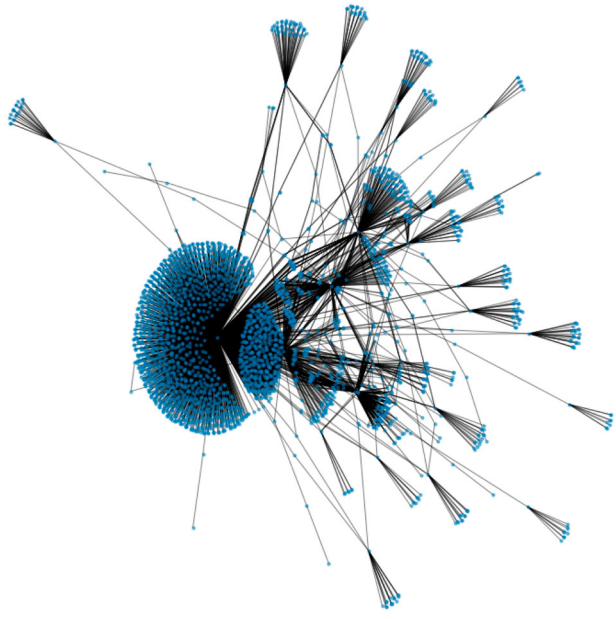


Figure 6. Achilles network visualisation.

Table 5. Number of entities.

Entity type	Number of entities
Company	3967
Country	61
Product	3396

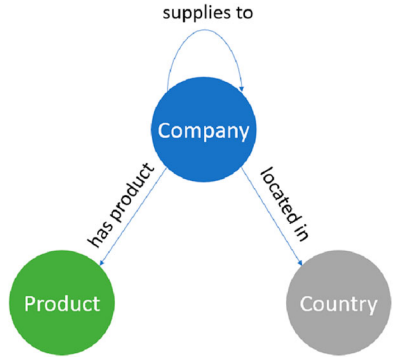


Figure 7. Proposed ontology.

The training and testing datasets are shown in Table 6. The number of positive and negative samples for each type of edge are shown. Similar to the automotive dataset, we also vary the number of negative samples for the test dataset in order to study imbalance.

The model performance is shown in Figure 8. 90% Confidence Interval for each performance measurement is displayed.

We discuss these results next.

Table 6. Achilles train and test dataset.

Edge type	Links	Negative samples	Ratio
		(a) Achilles Train Dataset	
Supplies to	2912	2912	1:1
Located in	2734	2734	1:1
Makes product	33,653	33,653	1:1
		(b) Achilles Test Dataset	
Supplies to	832	832/1664/2496/3328	1:1/2/3/4
Located in	781	781/1562/2343/3124	1:1/2/3/4
Makes product	9615	9615/19,230/28,845/38,460	1:1/2/3/4

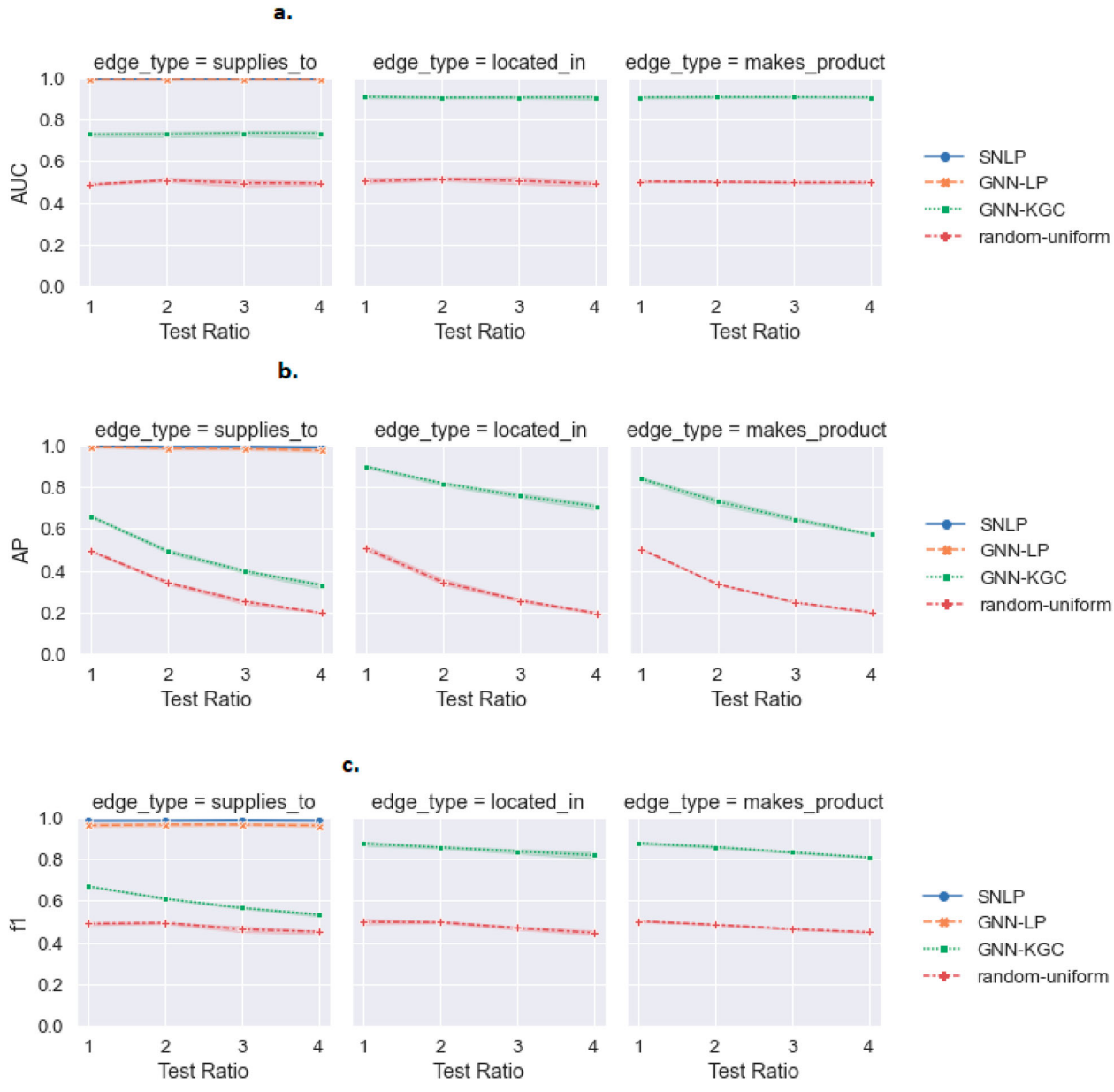


Figure 8. Performance on Achilles dataset. (a) Area Under curve across different test ratio, (b) average Precision across different test ratio and (c) f1-score across different test ratio.

6. Discussion

Similar insights can be derived from both experiments on Marklines and Achilles dataset.

As shown in Figures 5 and 8, the model can predict multiple types of hidden relationships, while SNLP and GNN-LP can only predict *supplies_to* edge type. This is

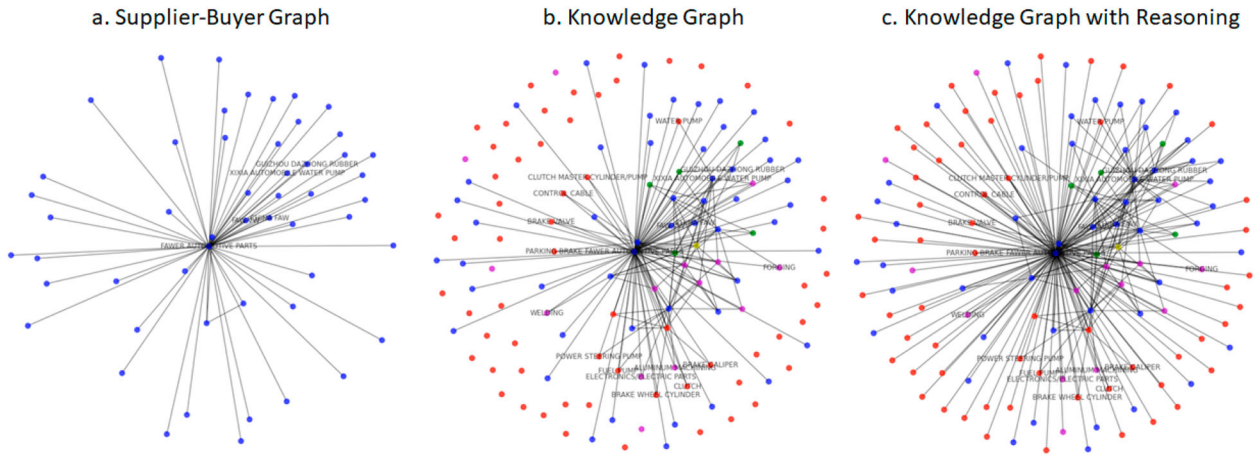


Figure 9. Illustration of how our model improves supply chain visibility. (a) Supplier–Buyer Graph shows the previous data representation where only companies are represented in the graph. (b) Knowledge Graph captures not only companies (blue colour) but also other entities such as products (red colour), countries (yellow colour), capabilities (purple colour) and certifications (green colour), each node coloured differently (c) Our model (Knowledge Graph with Reasoning) infers missing links between different entity types, allowing practitioners to query and identify unknown relationship risks.

due to the fundamental difference in both how we represent the data as a Knowledge Graph instead of a homogeneous graph, and due to the proposed GNN model, which can predict the existence of multiple edge types instead of just one.

To illustrate how our proposed model helps infer new information in SCRM, three graph visualisations are shown in Figure 9. The graphs show the different amounts of information that supply chain practitioners could progressively obtain. Figure 9(a) shows the current state of the art, where a homogeneous graph of who-supplies-whom is built and hidden links are inferred using methods proposed by Kosasih and Brintrup (2021) and Brintrup et al. (2018). Here we can answer queries such as *Which companies are Fawer Automotive Parts Ltd Co's suppliers?*.

Figure 9(b) shows the first part of our proposed approach; i.e. knowledge graph data representation, where more complex queries can be answered. For example: *Which suppliers of Fawer Automotive Parts Ltd Co produce brake valves for them?*. While the query can be posed, its answer is not guaranteed if data are missing, which is the case in this particular query.

This is where a link prediction (knowledge graph reasoning) approach helps fill in gaps in knowledge. Figure 9(c) shows how our model can infer hidden relationships in the graph as its density has increased. We can now identify the brake valve suppliers of Fawer Automotive Parts Ltd Co by traversing the new graph.

However, this new knowledge comes at the expense of a performance trade off, compared to algorithms developed specifically for single edge type prediction. For example, for the *supplies_to* edge type, our model sub-performs compared to SNLP and GNN-LP which had been tested on the same datasets. This is because GNN-LP and SNLP are trained on a singular edge type and thus are more specialised in their prediction. Developing heterogeneous GNN that works equally well to state of the art homogeneous GNN (such as GNN-LP) is an open machine learning research question that is being actively investigated in the wider literature. Nevertheless, performance of our method is still within acceptable means (e.g. 0.82 AUC on the automotive use case with a test ratio of 1) and thus is usable.

We can observe that across all edge types and models, the average precision and f1-score decrease as test ratio increases. This finding is consistent with the previous findings of Brintrup, Wang, and Tiwari (2017) and Kosasih and Brintrup (2021). This highlights the inherent problem of data imbalance with link prediction and KG reasoning in sparse graphs like supply chain. There are more negative than positive samples. Improving training of neural network models in view of data imbalance is an active area of research.

On the other hand, it appears that for the *supplies_to* edge type, precision decreases more severely in GNN-KGC compared to the more robust GNN-LP and SNLP. This is once more due to the nature of the GNN-KGC algorithm which attempts to balance all edge types. Whilst using GNN-KGC always improves practitioner knowledge, its efficacy needs further improvement in severely imbalanced cases. We further observe that the automotive use case yields superior results in all performance scores than the energy the use case. The AUC and AP of automotive use case are 0.82 and 0.72, respectively; whereas they are 0.73 and 0.66 in the energy use case

(both for test ratio 1). This can be attributed to differences in sample size. The largest edge type category in both datasets is that of *supplies_to*. This edge category alone in the energy use case is 9.4% of the size of automotive use case. There are two more entity types in the automotive use case. The combination of both these factors means a performance loss as both the dataset size and the ontology used to train the algorithm are smaller. However, the resulting performance loss for AUC and AP is 11% and 8%, respectively for test ratio 1, which is a relatively small difference, given a ten-fold discrepancy between data sizes in the two independent samples.

7. Conclusions

Modern supply chains contain hidden risks that could damage companies who are exposed to them. In this work, we developed a hybrid AI-based method to detect risk pertaining to hidden relationships between entities in the supply chain and showed how new knowledge can be inferred using Graph Neural Networks and Knowledge Graph Reasoning.

Our contribution fits into the body of supply chain risk management literature more broadly, and digital and AI-based supply chain risk surveillance more specifically. We address an important research gap in literature, characterised by the lack of techniques that can infer hidden information in existing data. As extant intelligence gathering methods such as supplier surveys are difficult to verify and rely on suppliers' willingness to share information, methods that can infer 'hidden' unknowns are valuable for risk management practitioners. To that extent, we improve upon previous AI-based approaches by developing a hybrid algorithm that can identify multiple types of risks, making it possible to create complex risk queries. Using two real-life datasets from the automotive and energy industries, we have shown that the approach allows practitioners to both ask complex queries beyond 'who-supplies-whom' and to fill in knowledge gaps by inferring multiple edge types.

Our results highlight both *technical* and *practical* improvements over state of the art as well as raising challenges.

While our current approach works for multiple types of risks, it can only handle single predicate logic statements at a time. However, in practice, risk queries would often be even more complex, combining several predicate logic statements from multiple types of risks together. For example, to answer queries such as '*Which company buys from a supplier that is located in Ayutthaya?*', one can conjugate two predicate logic statements *supplies_to (company A, company B)* $\wedge$ *is_located(company A, Ayutthaya)* and determine its satisfiability. A future extension to this approach is thus the exploration of multi-hop reasoning combining several triplets. However, this is a complex task, when there are unknown intermediate entities. The search space for such candidate entities could grow to as large as the number of entities in the graph. This is an active area of research within the wider machine learning literature (Ren and Leskovec 2020; Ren, Hu, and Leskovec 2020; Ren et al. 2021).

Another important limitation is that our model can only perform reasoning over entities that are present in the training dataset. Real supply chains, however, are dynamic systems where companies join and leave the system over time. Hence, there is a need to build a model that could work with new, previously unseen entities. We can explore inductive GNN models in the future, which is a class of techniques that can work with unseen entities without prior topological information. Building such an inductive machine learning method is another active area of research within the machine learning community (Teru, Denis, and Hamilton 2020) which we hope to investigate in the future.

Lastly, our model requires manual ontology engineering that needs to be carried out by domain experts. While this results in inherent explainability of the data representation, there is a scalability trade off as data complexity grows. Automated extraction of ontologies from large supply chain datasets helps address scalability issues (Khadir, Aliane, and Guessoum 2021).

From a practitioner standpoint, because the approach relies on AI and automated inference, practitioners need supply chain data to be in a digital and standardised format, that is free of errors. For adoption to be pursued, companies need to ensure that their IT systems support data standardisation via common schema and data dictionaries such that data from multiple sources could be brought together and analysed. As is common with many industrial AI applications, supply chain managers need to find appropriate ways to incorporate AI-based inference into business processes by placing appropriate checks and governance practices in place (Brintrup et al. 2022).

Moreover, as our model is a data-driven ML algorithm, it performs better with more data that is error free. Therefore, practitioners that wish to our approach will need to start from partially visible network data that is free of errors to the extent possible. The more data the practitioner has, the better the algorithm is set to perform. Practitioners may thus start experimenting with the algorithm using their internal knowledge of the network, and also consider collecting additional data on their industry manually, or through third-party information providers. A number of automated data collection approaches have also been proposed with open source code and training data available which may be utilised to

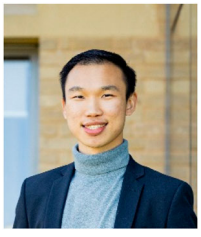
collect data in a given industrial domain from the world wide web (Wichmann et al. 2020).

Several authors have raised the urgency of explainability in industrial AI adoption. The neurosymbolic approach we have taken here helps improve explainability of AI in decision making process. Many state of the art AI algorithms are black box, where predictions are made without any explanation (Baryannis et al. 2019). Such a lack of transparency could reduce the trustworthiness from the respective human decision makers, hence preventing wider adoption of AI. Our approach inherently helps improve explainability by building in supply chain domain knowledge to our Knowledge Graph. Prediction is then treated as a reasoning task over different type of links. Using a neurosymbolic approach, we can answer risk queries from practitioners by first converting them to predicate logic statements and subsequently check their satisfiability using neural networks. Our work thus contributes to the growing literature that improves the trustworthiness of AI models to increase wider adoption of AI in the supply chain. However, causal interpretability of inferred links is non-trivial with the utilised GNN approach, and hence, the explainability of AI-based supply chain risk management needs improvement as this domain of inquiry grows in the future.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors



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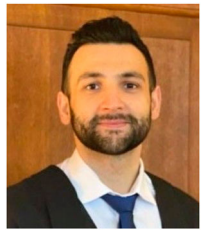
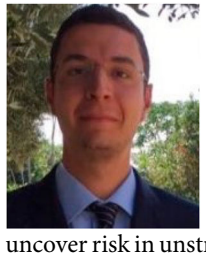
Nick Wildgoose is a qualified accountant, supply chain and risk professional and has held a variety of global financial, procurement and commercial positions in several industry sectors. Working for companies such as PWC, Linde Group, The Virgin Group, Zurich Insurance Group and DHL Resilience360. He has served on the Board of the Chartered Institute of Procurement and Supply which is the biggest procurement professional body in the world with over 100,000 members. He also served as a specialist advisor to the World Economic Forum on the topic of systemic supply chain risk and was Chairman of the Supply Chain Risk Leadership Council and still serves on their Board. He has spoken and written extensively on supply chain management.



Ajmal Aziz completed his MPhil at the University of Cambridge where he graduated with distinction. His research during his time at Cambridge focused on applying graph neural networks to supply chain knowledge graphs. He is currently a Solutions Architect at Databricks.



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Data availability statement

Due to the commercial nature of this research, supporting data is not available.

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